活菌的早期检测与分类

Early detection and classification of live bacteria using time-lapse coherent imaging and deep learning

时序相干成像与深度学习

Wang 等 · Light: Science & Applications 9, 118 · 2020

BL0028 · 面向微生物检测 introduction 与光学仪器研究

内容

检测问题与信号

活菌生长、全息图、时间信息

计算原理

拼接、角谱传播、复场差分

模型与证据

检测/分类网络、训练与独立盲测

结果与边界

全部主图、补充图表、时间与回收率

仪器研究支持

通量假设、可迁移思路、未证明事项

把“肉眼可见”提前到“微尺度可辨”

guidelines[11–13]. While the use of liquid growth media for the detection of fecal coliform bacteria provides high sensitivity and specificity, it requires at least 18 h for the final read-out. The use of solid agar plates is a relatively more cost-effective method and provides flexibility for the volume of the sample to be analysed, which can vary from 100 mL to several litres to enhance the sensitivity. However, this traditional culture-based detection method requires the colonies to grow to a certain macroscopic size for visibility, which often takes 24–48 h in the case of bacterial samples. Alternatively, molecular detection methods[14,15] based on, e.g., the amplification of nucleic acids, can reduce the assay time to a few hours, but they generally lack the sensitivity for detecting bacteria at very low concentrations, e.g., 1 colony forming unit (CFU) per 100–1000 mL, and are *not* capable of differentiating between live and dead microorganisms[16]. Furthermore, there is *no* EPA-approved nucleic acid-based analytical method[17] for detecting coliforms in water samples.

论文面对大体积、低浓度水样：传统平板计数要等菌落长到可见大小，文中以约24–48 h作背景。

目标信号是随时间出现并生长的菌落形态；输出为位置、计数及三类菌的类别。

这是2020年论文的背景判断，不把当年的审批状态当作当前法规结论。

原文截图 主文 PDF p.2, 引言：培养时间与检测问题

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.2, 引言：培养时间与检测问题
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研究对象与检测范围

suspensions spiked with coliform bacteria, including *E. coli*, *K. aerogenes*, and *K. pneumoniae*, and chlorine-stressed *E. coli*. A chromogenic agar medium designed for the specific detection and counting of *E. coli* and other coliform bacteria in food and water samples was used as a culture medium for specificity (see the "Methods" section for details). This chromogenic medium results in a blue colour for *E. coli* colonies and a mauve colour for the colonies of other coliform bacteria (e.g., *K. aerogenes* and *K. pneumoniae*). In addition, the medium inhibits the growth of different bacteria (e.g., *Bacillus subtilis*) or yields colourless colonies in the presence of other bacteria in the sample[27].

原文截图 主文 PDF p.3, Results：加标对象与选择性介质

验证对象为加标水样中的 E. coli、K. aerogenes、K. pneumoniae，并考察受氯应激的 E. coli。

生长介质本身带有选择性；不能把三类分类器视为任意微生物识别器。

本讲解分析光学测量与证据，不把样品处理图转换为培养或病原操作教程。

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.3, Results：加标对象与选择性介质
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Fig.1｜装置的完整定位

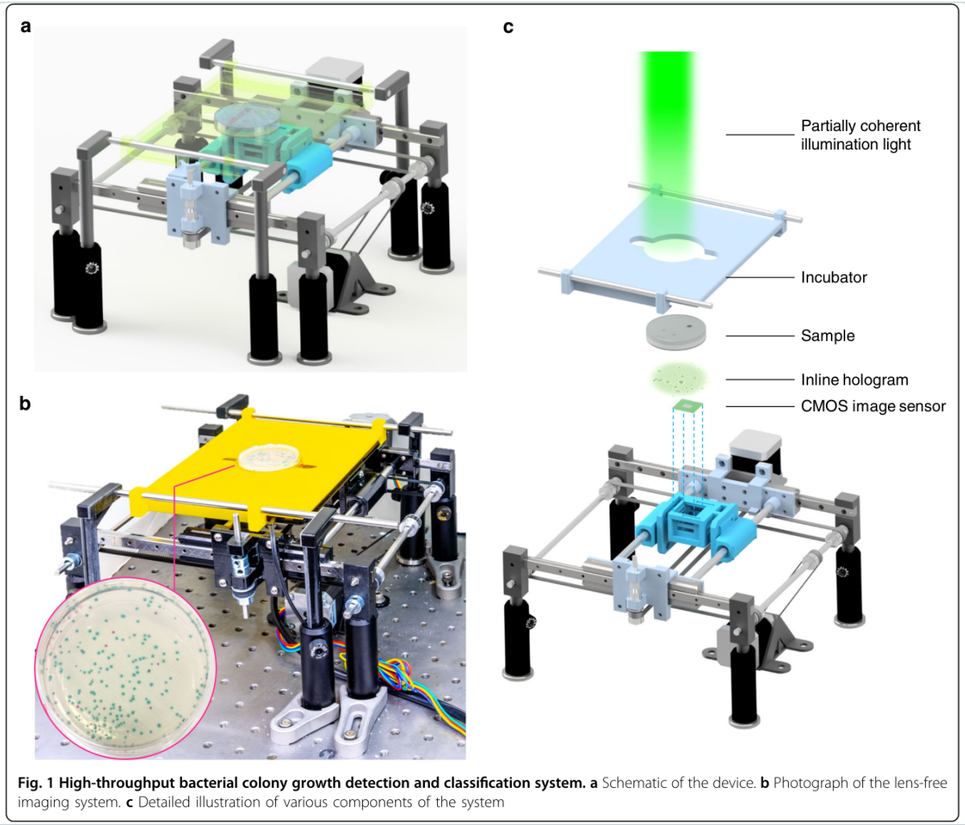


Fig. 1 High-throughput bacterial colony growth detection and classification system. a Schematic of the device. **b** Photograph of the lens-free imaging system. **c** Detailed illustration of various components of the system

原文截图 主文 PDF p.3, Fig.1 全图及原图注

a 机械与光学示意；b 实物；c 从照明、样品到传感器的光路。以下分别放大。

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扫描平台与视场

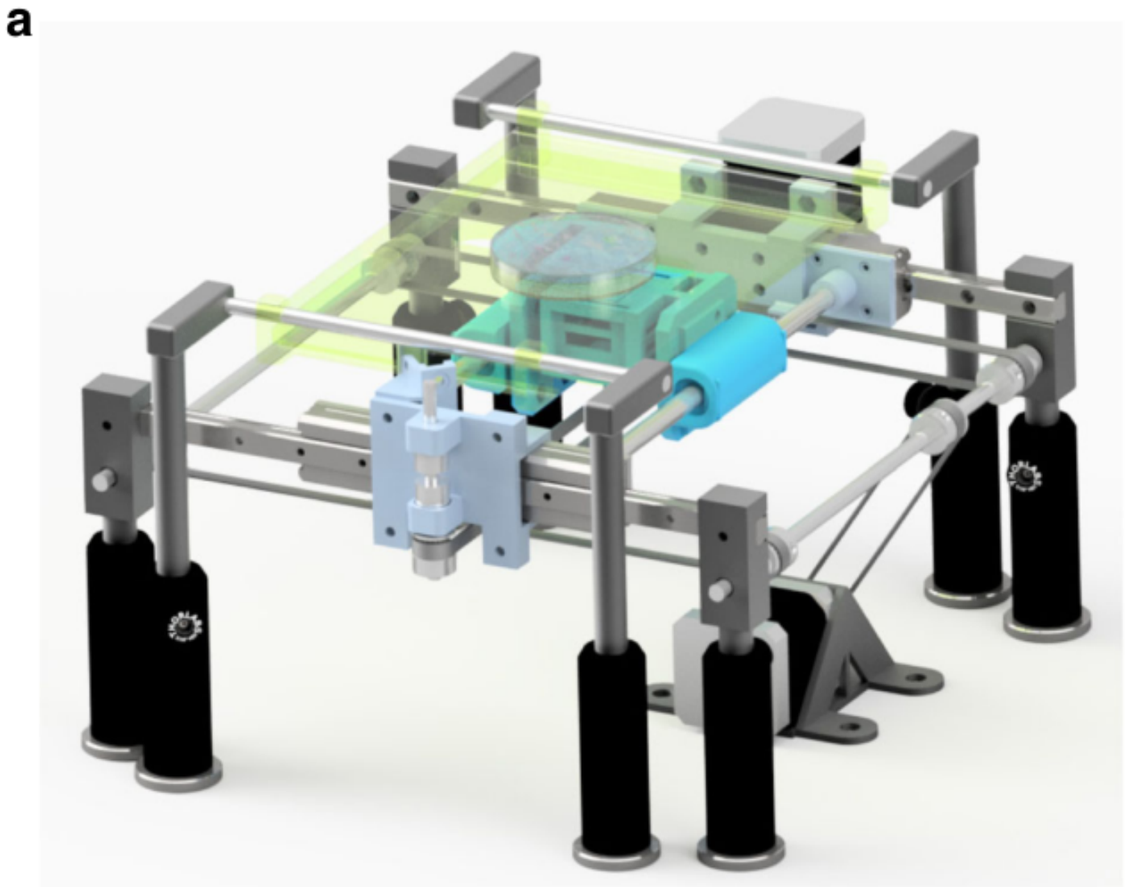


Fig.1a 显示二维移动机构。小传感器逐位置采集，软件把重叠图块拼成大视场。

输入是多个局部全息强度图；输出是同一时点的整板图像。

扫描并不增加单帧分辨率。扩大视场还要付出移动、曝光、传输与拼接时间。

原文截图 主文 PDF p.3, Fig.1a（对应完整子图/局部面板放大）

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.3, Fig.1a（对应完整子图/局部面板放大）
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实物图不能替代有效面积定义

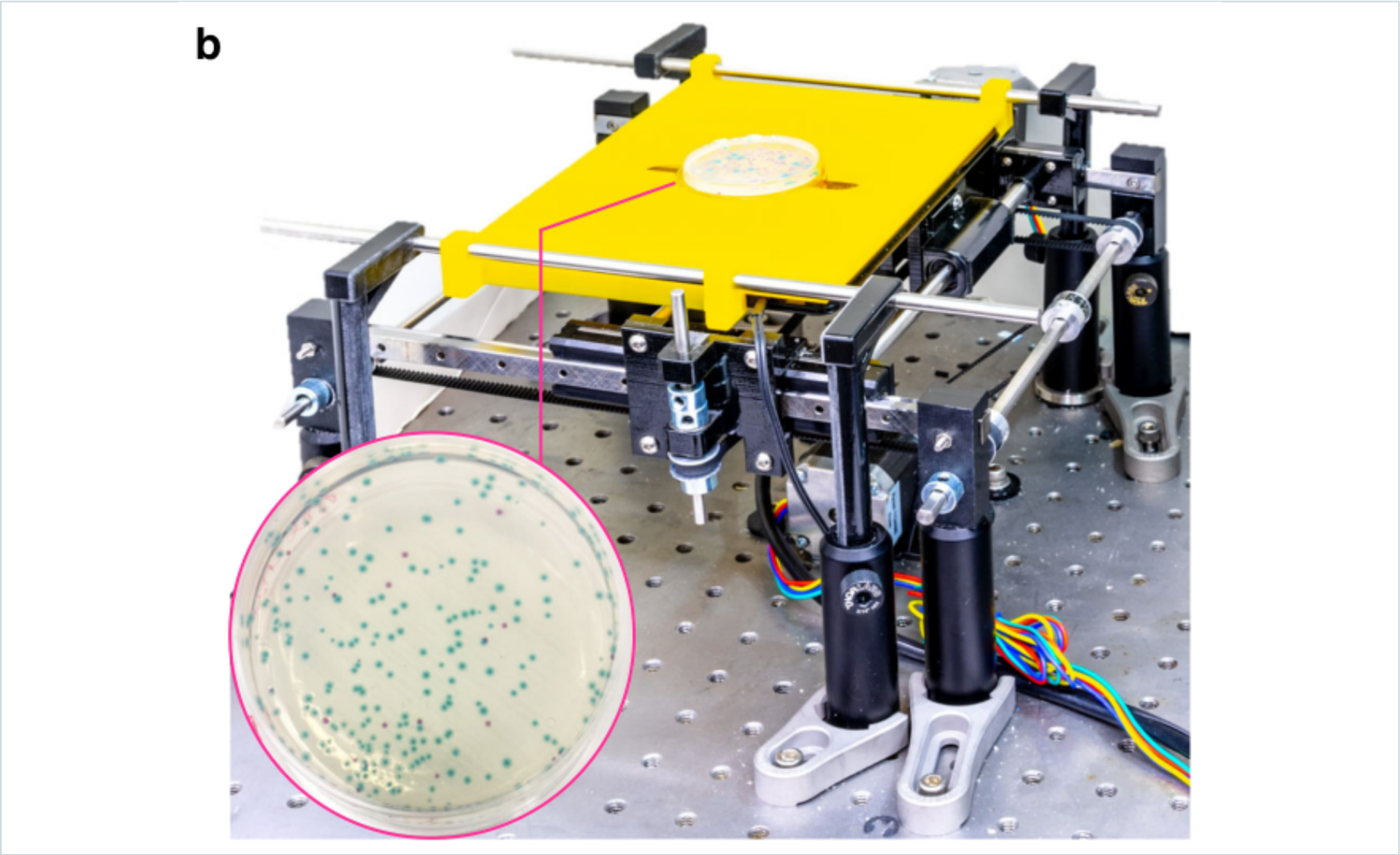


Fig.1b 表明装置与平板的机械组合，插图是培养后宏观平板。

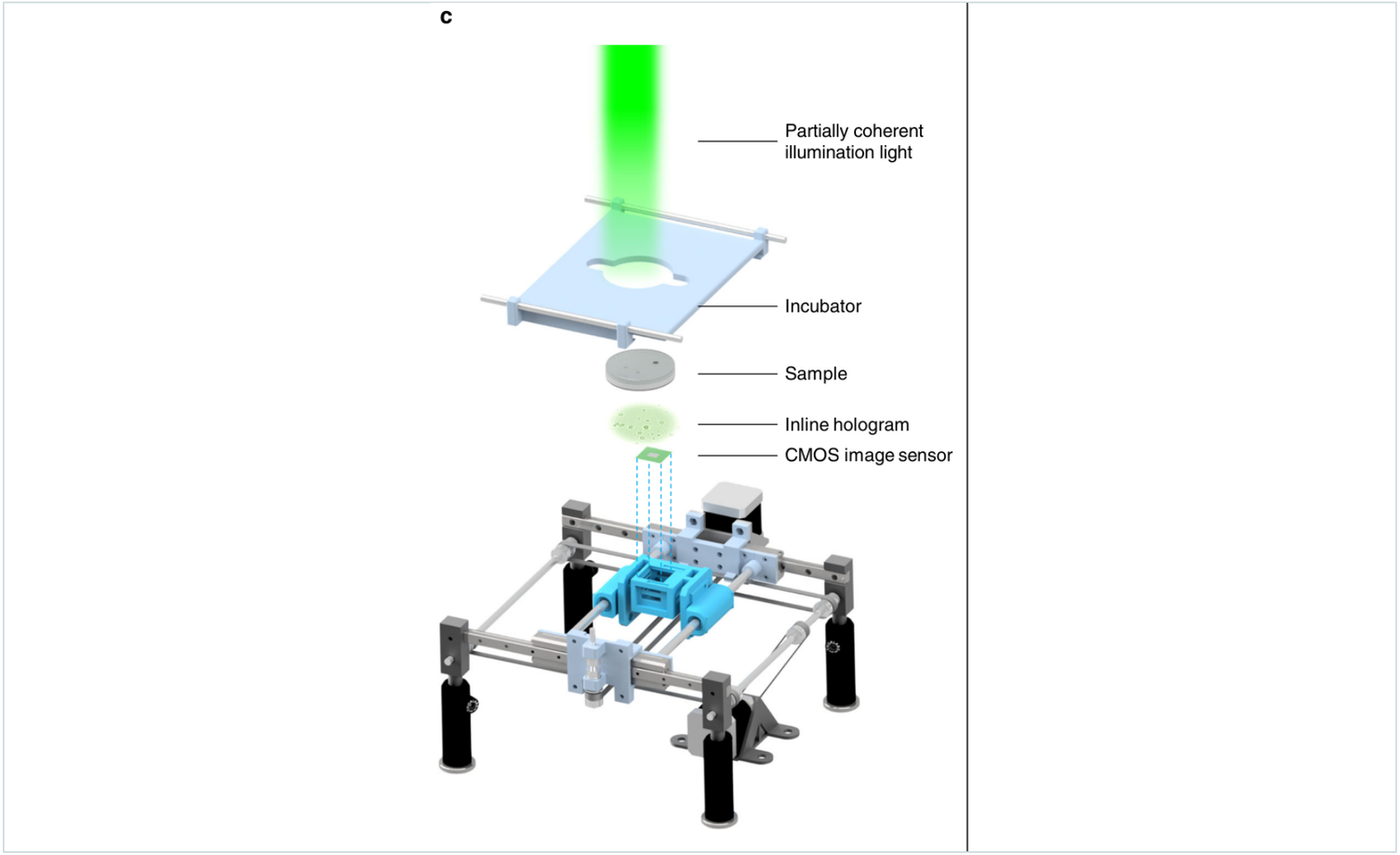
物理平板直径60 mm；后续真值计数仅使用中央50 mm圆，避开弯曲边缘伪影。

因此“扫描整板”与“经过验证的计数区域”应分别记录。

原文截图 主文 PDF p.3, Fig.1b（对应完整子图/局部面板放大）

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.3, Fig.1b（对应完整子图/局部面板放大）；另据：主文p.260mm；p.15中央50mm
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同轴全息测量的信号链



部分相干光穿过样品；未散射参考光与散射光在CMOS上叠加，形成同轴全息强度图。

传感器记录 $I=|R+O|^2$，其中R为参考场、O为物场。复振幅和相位并非各自直接测得。

数字回传给出复数重建；输入传感器强度与输出重建相位不能混称。

原文截图 主文 PDF p.3, Fig.1c（对应完整子图/局部面板放大）

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.3, Fig.1c（对应完整子图/局部面板放大）
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弱物体近似与孪生像边界

Colony candidate selection by differential analysis

When a new frame was acquired at time t, it was cross-registered to the previous frame at time $t-1$ and then digitally back-propagated to the sample plane[44,45] to obtain the complex light field

$$\widetilde{B}_t = \mathrm{P}(F_t, \mathbf{z}) \tag{3}$$

where F_t is the frame at time $\mathbf{t}$, $\mathbf{z}$ is a surface normal vector of the sample plane obtained by digital autofocusing[46] at 50 randomly spaced positions, and P denotes the angular spectrum-based back-propagation operation[44,45], which can be calculated by multiplying the spatial Fourier transform of the input signal and the following transfer function

$$H_k(\nu_x, \nu_y) = \begin{cases} \exp\left[-j \cdot 2\pi \frac{n \cdot z}{\lambda} \sqrt{1 - \left(\frac{\lambda}{n}\nu_x\right)^2 - \left(\frac{\lambda}{n}\nu_y\right)^2}\right] & \left(\nu_x^2 + \nu_y^2 \le \left(\frac{n}{\lambda}\right)^2\right) \\ 0 & \text{otherwise} \end{cases}$$

where n is the refractive index of the medium, λ is the illumination wavelength, and ν_x and ν_y are the spatial frequencies. This operation was followed by an inverse 2D Fourier transform. The resulting complex-valued reconstruction provides both the amplitude and phase images of the illuminated objects. To accommodate the large FOV of a stitched frame (36,000 × 36,000 pixels), digital back-propagation was performed with 2048 × 2048-pixel blocks, which were then merged together.

Four consecutive frames were taken, i.e., from $t-3$ to t, and a differential image was calculated defined by

原文截图 主文 PDF p.14, 式(3)、角谱传递函数及符号

教学解释：$I=|R|^2+R^*O+RO^*+|O|^2$。若物场相对参考场较弱，$|O|^2$项较小，交叉项携带结构信息。

单幅强度不足以唯一确定任意复场；同轴结构仍可能存在孪生像与背景伪影。

作者实现角谱回传与数字聚焦，并未在本文实现多高度迭代相位恢复。上述展开是物理解释，不是新增实测结果。

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角谱传播｜输入、算子、输出

Colony candidate selection by differential analysis

When a new frame was acquired at time t, it was cross-registered to the previous frame at time $t-1$ and then digitally back-propagated to the sample plane[44,45] to obtain the complex light field

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Four consecutive frames were taken, i.e., from $t-3$ to t, and a differential image was calculated defined by

式(3)：$B_t=P(F_t,z)$。先对配准后的整帧F_t做二维FFT，在频域乘H，再作IFFT。

H的指数给不同空间频率施加传播相位；超出传播频率圆的分量置零。

输出B_t为复数图，分为幅度$|B_t|$与相位$\arg(B_t)$，供时间差分和网络使用。

原文截图 主文 PDF p.14, 式(3)、角谱传递函数及符号

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角谱传播｜符号与空间尺度

Colony candidate selection by differential analysis

When a new frame was acquired at time t, it was cross-registered to the previous frame at time $t-1$ and then digitally back-propagated to the sample plane[44,45] to obtain the complex light field

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Four consecutive frames were taken, i.e., from $t-3$ to t, and a differential image was calculated defined by

原文截图 主文 PDF p.14, 式(3)、角谱传递函数及符号

n为介质折射率；λ为照明波长；ν_x、ν_y为空间频率。λ与传播距离同长度单位，ν为逆长度，n无量纲。

作者在50个随机位置数字聚焦得到表面参数；36,000×36,000像素整帧分成2048×2048块回传。

分块缓解计算量；块间拼接、错误z、边界与介质不均匀仍可能产生伪影。相位不能直接解释成单细胞质量。

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角谱传递函数｜原式放大

$$\widetilde{B}_t = \mathrm{P}(F_t, \mathbf{z}) \tag{3}$$

where F_t is the frame at time **t**, **z** is a surface normal vector of the sample plane obtained by digital autofocusing[46] at 50 randomly spaced positions, and P denotes the angular spectrum-based back-propagation operation[44,45], which can be calculated by multiplying the spatial Fourier transform of the input signal and the following transfer function

$$H_k(\nu_x, \nu_y) = \begin{cases} \exp\left[-j \cdot 2\pi \frac{n \cdot z}{\lambda} \sqrt{1 - \left(\frac{\lambda}{n}\nu_x\right)^2 - \left(\frac{\lambda}{n}\nu_y\right)^2}\right] & \left(\nu_x^2 + \nu_y^2 \le \left(\frac{n}{\lambda}\right)^2\right) \\ 0 & \text{otherwise} \end{cases}$$

P通过频域传播核H作用于输入。指数含−j、2π、n·z/λ以及空间频率相关根号项；j是虚数单位。

条件$\nu_x^2+\nu_y^2 \le (n/\lambda)^2$定义传播波频率区域，区域外在作者实现中置零。

负号对应本文回传播约定。相机强度起点、数字聚焦与弱物体限制仍按前页条件解释。

原文截图 主文 PDF p.14, 式(3)及完整角谱传递函数放大

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.14, 式(3)及完整角谱传递函数放大
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角谱传播｜一个频率的教学算例

$$\widetilde{B}_t = \mathrm{P}(F_t, \mathbf{z}) \tag{3}$$

where F_t is the frame at time $\mathbf{t}$, $\mathbf{z}$ is a surface normal vector of the sample plane obtained by digital autofocusing[46] at 50 randomly spaced positions, and P denotes the angular spectrum-based back-propagation operation[44,45], which can be calculated by multiplying the spatial Fourier transform of the input signal and the following transfer function

$$H_k(\nu_x, \nu_y) = \begin{cases} \exp\left[-j \cdot 2\pi \frac{n \cdot z}{\lambda} \sqrt{1 - \left(\frac{\lambda}{n}\nu_x\right)^2 - \left(\frac{\lambda}{n}\nu_y\right)^2}\right] & \left(\nu_x^2 + \nu_y^2 \le \left(\frac{n}{\lambda}\right)^2\right) \\ 0 & \text{otherwise} \end{cases}$$

非实测：取局部平面、标量传播距离z的教学近似。原文将z描述为表面法向向量，本例不替其给出真实几何。

设n=1、λ=0.532 μm、z=λ/2，ν_x=0.6/λ、ν_y=0。根号项$\sqrt{(1-0.6^2)}$=0.8，频率在传播区域内，但超过本机像素采样频带，详见下一页。

H=exp(−j・2π・0.5・0.8)≈−0.809−0.588j。输入频谱系数1+0j，输出同H：幅值1，相位−0.8π；再IFFT得到该空间模式。

原文截图 主文 PDF p.14, 式(3)及完整角谱传递函数放大

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.14, 式(3)及完整角谱传递函数放大
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传播频率与相机可采样频带

$$\widetilde{B}_t = \mathrm{P}(F_t, \mathbf{z}) \tag{3}$$

where F_t is the frame at time $\mathbf{t}$, $\mathbf{z}$ is a surface normal vector of the sample plane obtained by digital autofocusing[46] at 50 randomly spaced positions, and P denotes the angular spectrum-based back-propagation operation[44,45], which can be calculated by multiplying the spatial Fourier transform of the input signal and the following transfer function

$$H_k(\nu_x, \nu_y) = \begin{cases} \exp\left[-j \cdot 2\pi \frac{n \cdot z}{\lambda} \sqrt{1 - \left(\frac{\lambda}{n}\nu_x\right)^2 - \left(\frac{\lambda}{n}\nu_y\right)^2}\right] & \left(\nu_x^2 + \nu_y^2 \le \left(\frac{n}{\lambda}\right)^2\right) \\ 0 & \text{otherwise} \end{cases}$$

上页仅演示连续传播模型。其$\nu_x \approx 1.128\ \mu m^{-1}$；本文像素间距p=1.67 μm，对应Nyquist频率$1/(2p) \approx 0.299\ \mu m^{-1}$。上例不能当本机直接采到的细节。

在带内的教学例：$\nu_x=\nu_y=0$，n=1，取$z=\lambda/4=0.133\ \mu m$，则$H=\exp(-j\pi/2)=-j$。单位频谱系数变为−j，幅值仍为1。

传播波截止和传感器采样截止不同；数字回传不会自动恢复未采集的高频。此处z仍为局部平面标量教学距离，非实测样品厚度。

原文截图 主文 PDF p.14, 式(3)及完整角谱传递函数放大

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.14, 式(3)及完整角谱传递函数放大；另据：主文p.13 像素尺寸；教学采样计算
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拼接｜相位相关估计局部平移

$$\vec{p}_{AB} = (\Delta x, \Delta y) = \arg\max_{(x,y)} \mathcal{F}^{-1}\left\{\frac{\mathcal{F}\{A\} \cdot \mathcal{F}\{B\}^*}{|\mathcal{F}\{A\} \cdot \mathcal{F}\{B\}^*|}\right\} \quad (2)$$

where $\mathcal{F}$is the Fourier transform operator and $\mathcal{F}^{-1}$ is the inverse Fourier transform operator. The optimal

式(2)以两块重叠图A、B为输入：FFT后做一个频谱乘另一个的共轭，再除以模长。

IFFT把一致的平移相位变成相关峰；最大值的位置给出Δx、Δy，单位为像素。

归一化减弱整体幅值差影响，但无纹理、重复纹理、低信噪比或非平移变形仍会使峰不可信。

原文截图 主文 PDF p.14, 式(2)与傅里叶算子

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相位相关｜四点教学算例

$$\vec{p}_{AB} = (\Delta x, \Delta y) = \arg\max_{(x,y)} \mathcal{F}^{-1}\left\{\frac{\mathcal{F}\{A\} \cdot \mathcal{F}\{B\}^*}{|\mathcal{F}\{A\} \cdot \mathcal{F}\{B\}^*|}\right\} \quad (2)$$

where $\mathcal{F}$ is the Fourier transform operator and $\mathcal{F}^{-1}$ is the inverse Fourier transform operator. The optimal

教学算例，非原文测量。令A=[0,1,0,0]，B=[0,0,1,0]，B比A向右移1格。

取式(2)的F(A)F(B)*归一化并IFFT，得到[0,0,0,1]。峰索引3在循环坐标表示−1。

因此把B向左平移1像素即可对齐A。若交换A/B，位移符号反转；实现必须统一“谁对齐谁”。

原文截图 主文 PDF p.14, 式(2)与傅里叶算子

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.14, 式(2)与傅里叶算子
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拼接｜全局位置协调局部误差

Image stitching to obtain the image of the entire plate area

Following the acquisition of holographic images using the multi-threading approach, all the images within a tile-scan of the whole Petri dish per wavelength were merged into a single full-FOV image. During a tile scan, the images were acquired with ~30% overlap on each side of the image to calculate the relative image shifts against each other. For each image, the relative shifts against all four of the neighbouring images were calculated using a phase correlation[43] method, followed by an optimization step that minimized an object function, as defined by

$$\arg\min_{T_{VF}} \sum_{A\in V\setminus\{F\}} \left(\sum_{B\in V\setminus\{F\}} \left\| \vec{t}_{AF} - \vec{t}_{BF} - \vec{p}_{AB} \right\|^2 \right) \qquad (1)$$

where V is the set of all tile images, $F \in V$ is a fixed image, e.g., the image captured at the centre of the sample Petri dish, $\vec{t}_{AB}$ stands for the relative position of image A with respect to image B, and $\vec{p}_{AB}$ is the local shift between images A and B, calculated by the phase correlation method using the overlapping regions of the two neighbouring images, which can be formulated as

式(1)把每块相对固定参考F的位置作为未知量，最小化所有邻块位移残差平方和。

局部p_AB来自相位相关；全局t_AF是优化后的位置。固定参考消除整体平移，须有连通图及锚点关联边或等效绝对约束。

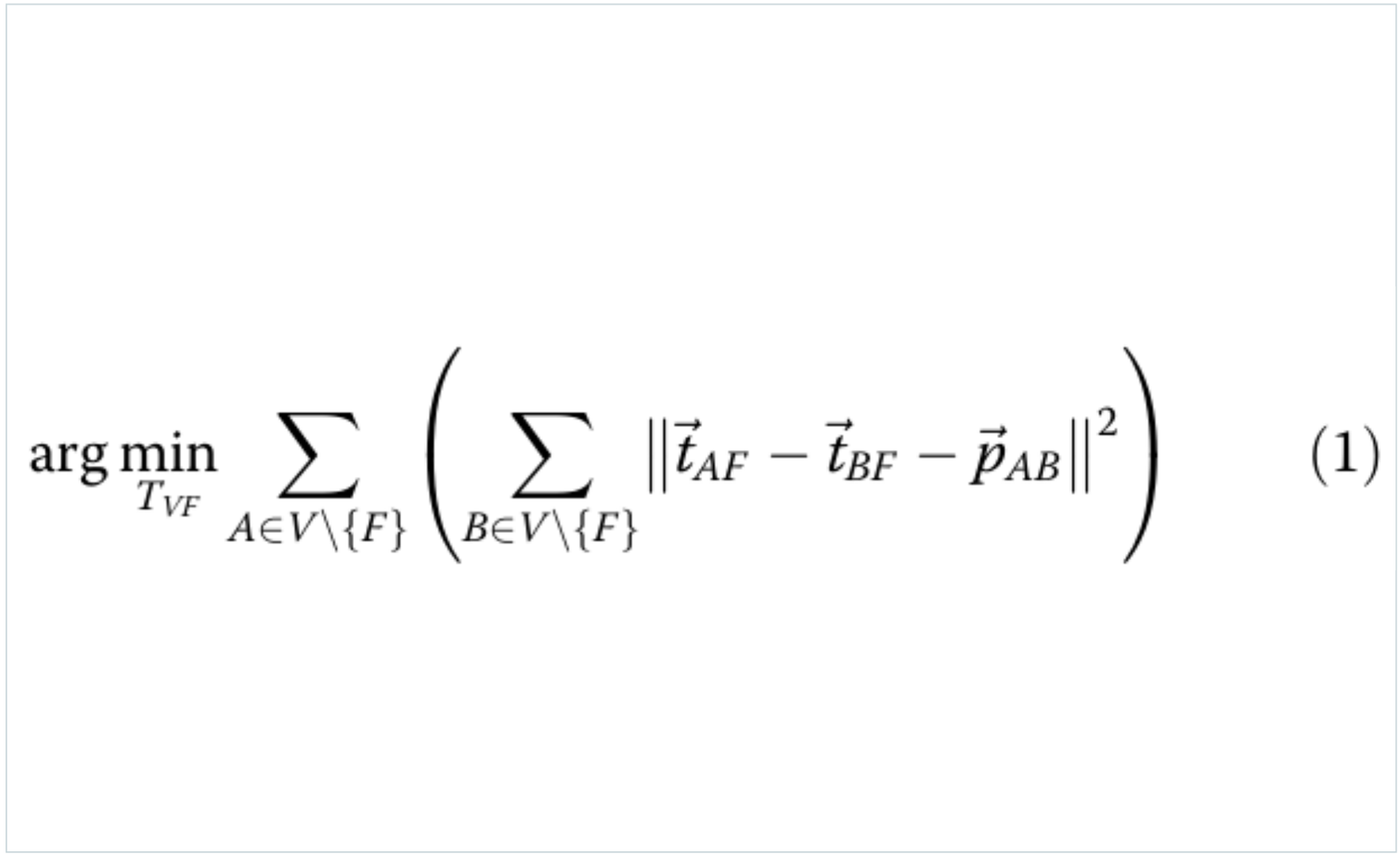
作者剔除相关系数<0.3的局部位移，随后线性融合，整帧均值归一到50并存为8-bit。

印刷式两重求和均排除F；若仅含这些差值，非F位置共同平移不改残差。此处不推断作者实现。

原文截图 主文 PDF p.14, 式(1)与位移符号

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三图块最小二乘｜完整演算



$$\arg\min_{T_{VF}} \sum_{A\in V\setminus\{F\}} \left(\sum_{B\in V\setminus\{F\}} \left\| \vec{t}_{AF} - \vec{t}_{BF} - \vec{p}_{AB} \right\|^2 \right) \qquad (1)$$

教学算例，非测量。固定x_A=0；估计B−A=100、C−B=100、C−A=203像素。

最小化$(b-100)^2+(c-b-100)^2+(c-203)^2$。求导得2b−c=0、2c−b=303。

解b=101、c=202像素。残差为1、1、−1，总平方误差3；逐块累积b=100、c=200则误差9。

本例图连通，B−A、C−A显式连接锚点A；这是可唯一求解的教学约束，并非按印刷式(1)求和范围逐字代入。

原文截图 主文 PDF p.14, 完整式(1)放大；符号见前页

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.14, 完整式(1)放大；符号见前页
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时间差分｜四帧形成三个间隔

Four consecutive frames were taken, i.e., from $t-3$ to t, and a differential image was calculated defined by

$$D_t = \mathrm{HP}\left[\mathrm{LP}\left(\frac{1}{3}\sum_{\tau=t-2}^{t}\left|\tilde{B}_\tau - \tilde{B}_{\tau-1}\right|\right)\right] \tag{4}$$

where D_t is the differential image at time t, $\tilde{B}_t$ represents the complex light field obtained by back-propagating frame t, and LP and HP represent low-pass and high-pass image filtering, respectively. The HP filter removes the differential signal from a slowly varying background (unwanted term), and the LP filter removes

式(4)使用t−3至t共四幅复场，先算三个相邻复场差的模，再取平均。

计算次序：复数相减→取绝对值→三项平均→LP低通→HP高通。不是原始强度相减，也不是先对相位单独求导。

LP抑制高频噪声；HP压制缓变空间背景。变化还可能来自气泡和表面运动，因此后续需要检测网络。

原文截图 主文 PDF p.14, 式(4)：复场时间差分、LP与HP

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.14, 式(4)：复场时间差分、LP与HP
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复场差分｜模与强度差不能互换

Four consecutive frames were taken, i.e., from $t-3$ to t, and a differential image was calculated defined by

$$D_t = \mathrm{HP}\left[\mathrm{LP}\left(\frac{1}{3}\sum_{\tau=t-2}^{t}\left|\tilde{B}_\tau - \tilde{B}_{\tau-1}\right|\right)\right] \tag{4}$$

where D_t is the differential image at time t, $\tilde{B}_t$ represents the complex light field obtained by back-propagating frame t, and LP and HP represent low-pass and high-pass image filtering, respectively. The HP filter removes the differential signal from a slowly varying background (unwanted term), and the LP filter removes

教学算例，非实测。某像素四帧为1、1+i、1+2i、2+2i，单位采用归一化场幅。

三个复数差依次i、i、1；模均为1，滤波前平均=1。

对应强度为1、2、5、8，强度差平均=(1+3+3)/3=7/3，数值不同。单像素例不能计算空间LP/HP；滤波还需邻域。

原文截图 主文 PDF p.14, 式(4)：复场时间差分、LP与HP

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.14, 式(4)：复场时间差分、LP与HP
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时间窗与物理时间跨度

validation dataset to finalize our network models and achieved a TPR of ~95% and a PPV of ~95% once the network converged, which took ~4 h of training time. Examples of the training loss and detection accuracy curves are shown in Supplementary Fig. S1.

The second step further classifies the species of the detected colonies with a classification DNN model following a similar network architecture. To accommodate the different growth rates of bacterial colonies, we used a longer time window in this classification neural network, containing 8 consecutive frames ($8 \times 0.5 = 4$ h) for each sub-ROI. Since the bacterial growth detection network uses a shorter running time window of 2 h, there is a natural 2-h time delay between the successful detection of a growing colony and the classification of its species. The network was trained with 7919 growing colonies, which contained 3362 *E. coli*, 1880 *K. aerogenes*, and 2677 *K. pneumoniae* colonies, and it was validated with 340 *E. coli*, 205 *K. aerogenes*, and 988 *K. pneumoniae* colonies from 6 independent plates and reached a validation classification accuracy of ~89% for *E. coli*, ~95% for *K. aerogenes*, and ~98% for *K. pneumoniae* when the network model converged (Supplementary Fig. S2).

After these network models were finalized through the training and validation data, we tested their generalization capabilities with an additional set of experiments that were never seen by the networks before; the results of these blind tests are detailed next.

采集间隔为30 min。作者称4帧窗口“2 h”、8帧窗口“4 h”，沿用帧数×周期的窗口命名。

四个采样时刻首尾实际相隔3×0.5=1.5 h；八帧首尾相隔7×0.5=3.5 h。

分类比检测多4帧，即再等待2 h。写时间边界时应区分采样数、首尾跨度与首次可输出的时刻。

原文截图 主文 PDF p.5, 分类训练/验证与时间窗

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Fig.2｜完整测量与计算流程

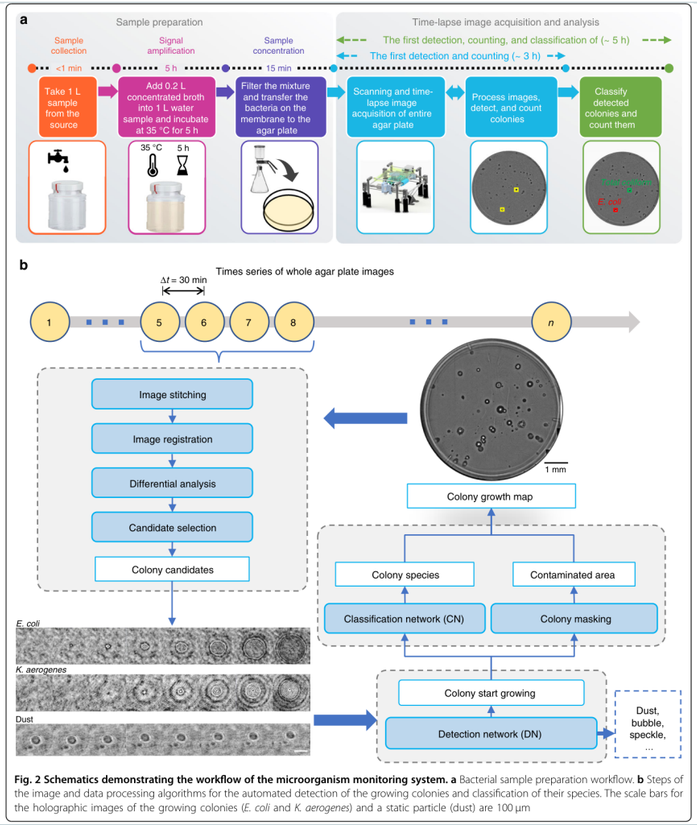


Fig. 2 Schematics demonstrating the workflow of the microorganism monitoring system. a Bacterial sample preparation workflow. **b** Steps of the image and data processing algorithms for the automated detection of the growing colonies and classification of their species. The scale bars for the holographic images of the growing colonies (*E. coli* and *K. aerogenes*) and a static particle (dust) are 100 μm

原文截图 主文 PDF p.4, Fig.2 全图及原图注

a 显示样品到输出的时间链；b 显示先找增长、再分类的处理链。原图只作研究解释，不作操作指南。

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.4, Fig.2 全图及原图注
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总测试时间包含样品环节

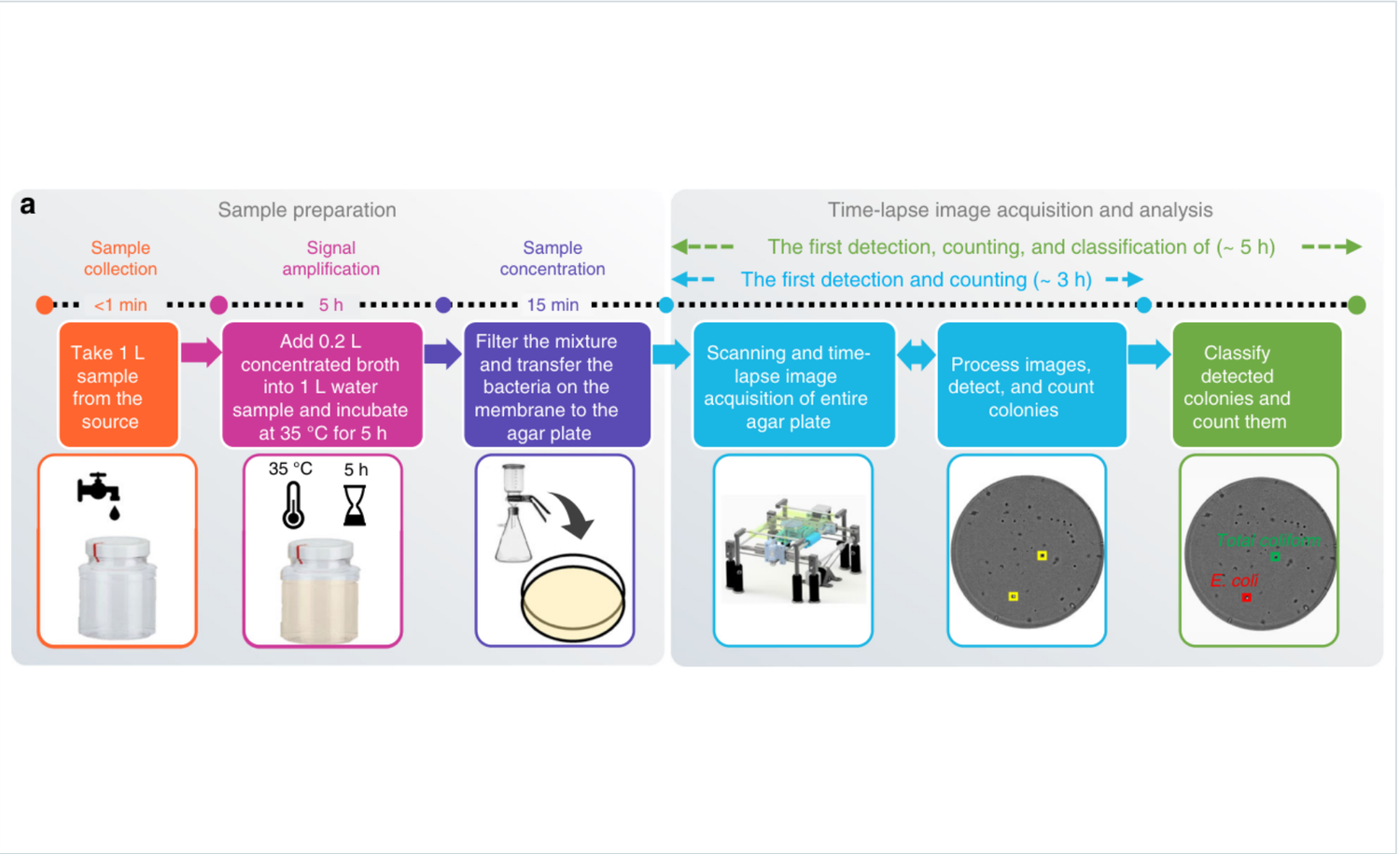


Fig.2a把液相预培养、浓缩/转移、扫描与判断串在一起；约1 CFU/L、≤9 h的结果依赖预培养。

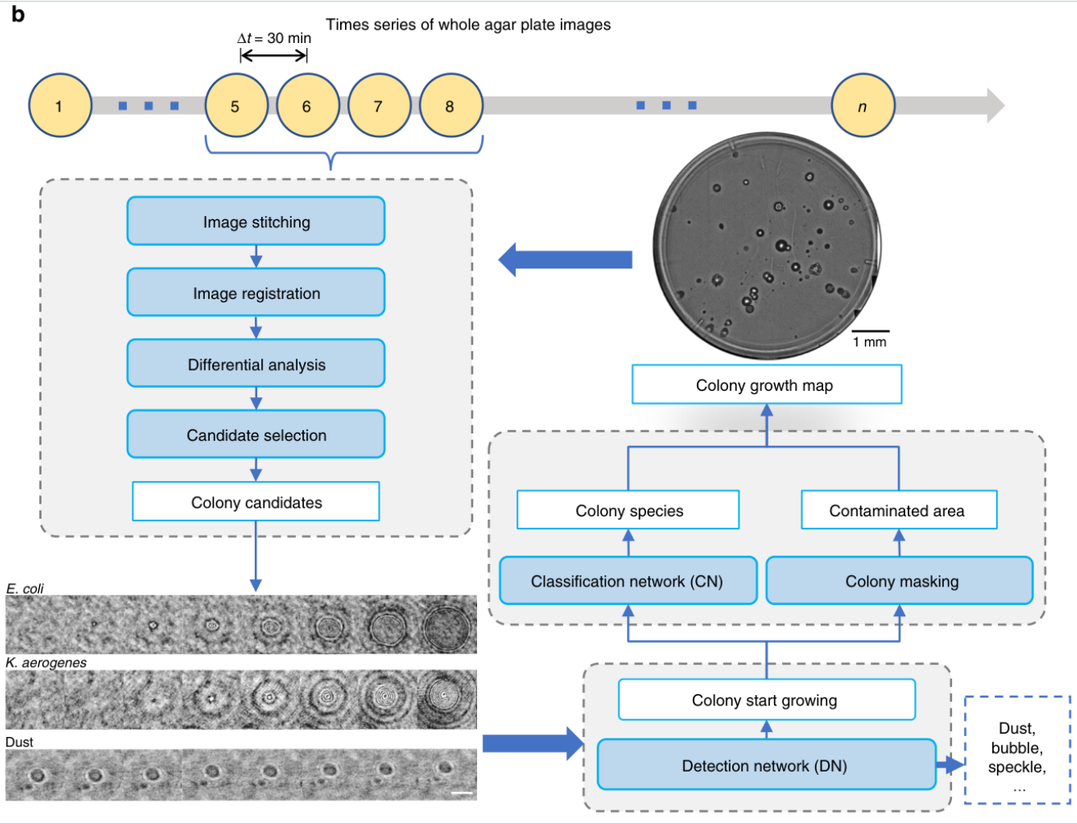
图中“首次检测”与“首次检测并分类”分别在成像后约3 h和约5 h，并非所有菌落均已检出。

样品环节改变可检测CFU数量与回收率；不能把总时间写成纯算法运行时间。

原文截图 主文 PDF p.4, Fig.2a（对应完整子图/局部面板放大）

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.4, Fig.2a（对应完整子图/局部面板放大）
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Fig.2b｜两级网络的完整路径



原文截图 主文 PDF p.4, Fig.2b（对应完整子图/局部面板放大）

差分候选包含菌落与动态杂质。检测网络先筛选，分类网络再处理已检出的菌落；漏检不进入分类。

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.4, Fig.2b（对应完整子图/局部面板放大）
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候选筛选与背景干扰

the high-frequency noise-introduced spatial patterns. The LP and HP filter kernels were empirically set to 5 and 100, respectively.

Following the differential image calculation, we selected regions in the differential image with >50 connective pixels that are above an intensity threshold, which was empirically set to 12. These regions are marked as colony candidates, as they give a differential signal over a period of time (covering four consecutive frames). However, some of the differential signals come from nonbacterial objects, such as a water bubble or surface movement of the agar itself. Therefore, we also used two DNNs to select the true candidates and classify their species.

拼接与跨时刻配准在差分之前；否则整个背景的移动也会形成变化。

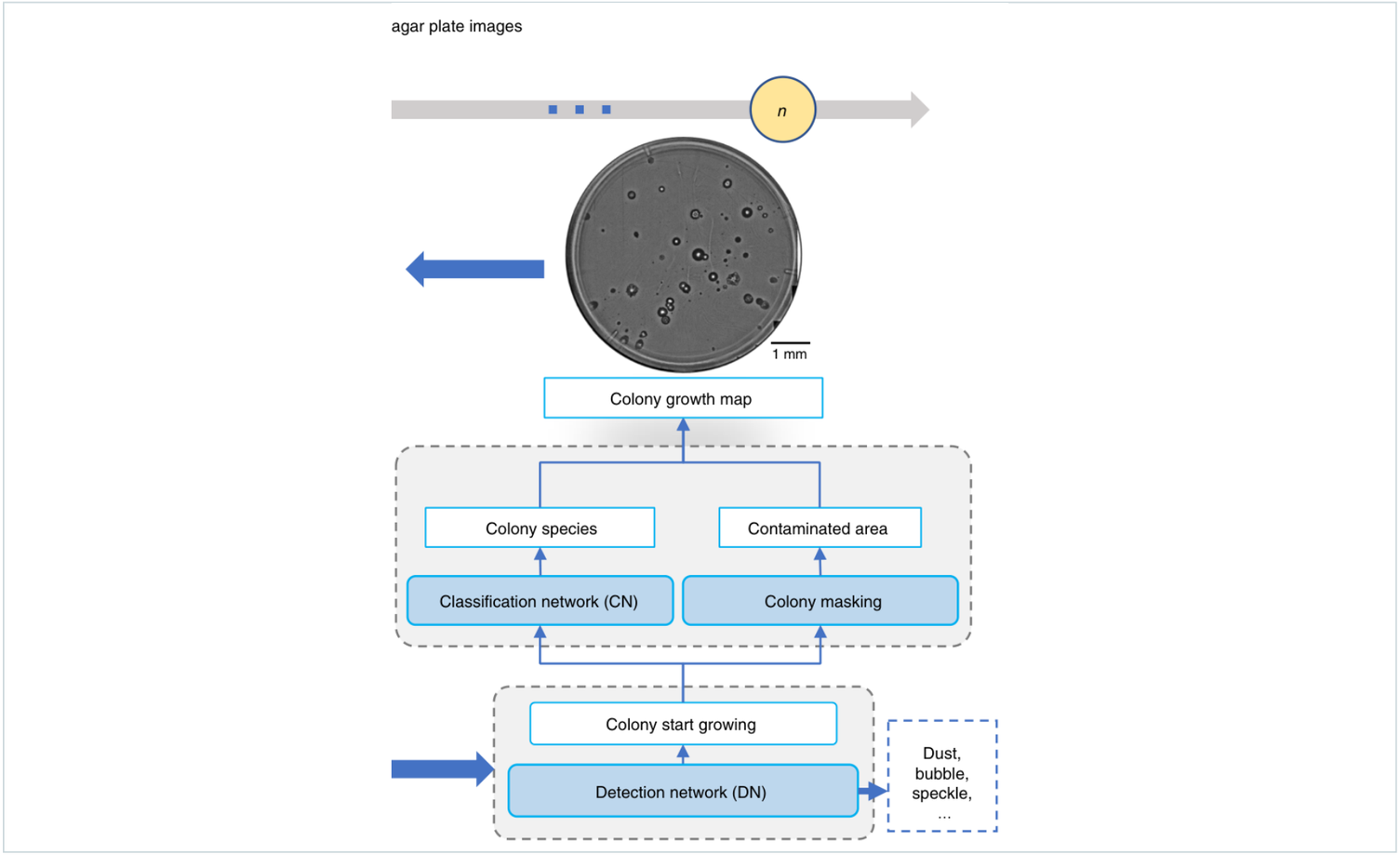
作者在差分图中用强度阈值12与>50连通像素选候选，LP/HP核分别设为5/100。

这些是该数据链上的经验参数，依赖归一化和采样尺度，不是通用菌落的物理阈值。

原文截图 主文 PDF p.15, 差分滤波核与候选阈值

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.15, 差分滤波核与候选阈值
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检测与分类的不同输出



检测网络回答候选是否为生长菌落，排除灰尘、气泡、散斑等。

分类网络只接收已检测菌落，给出三个已知类别之一；掩膜与类别共同形成生长图。

检测阳性分数不是物种概率；闭集分类也没有在本文验证“未知种拒绝”能力。

原文截图 主文 PDF p.4, Fig.2b 右侧检测/分类链

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.4, Fig.2b 右侧检测/分类链
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光学采集参数与采样

computer through a universal serial bus (USB) 3.0 interface and was software-triggered within the same C++ program. The exposure time at each scanning position was precalibrated according to the intensity distribution of the illumination light and ranged from 4 to 167 ms. The images were saved as 8-bit bitmap files for further processing.

传感器像素1.67 μm、有效面积6.4×4.6 mm^2；论文采集用532 nm照明。

曝光按位置预标定，范围4–167 ms；传感器和照明均受程序控制。

曝光时间影响动态范围/信噪比，像素间距与重建共同限制可见细节；不能由“无透镜”推出无限分辨率。

原文截图 主文 PDF p.13, Methods：传感器、曝光与扫描

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 主文 PDF p.13, Methods：传感器、曝光与扫描；另据：主文p.13–14 Methods光学模块/采集
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传感器尺寸与全息采集模块

Design of the high-throughput time-resolved microorganism monitoring platform

Our platform consists of five modules: (1) a holographic imaging system, (2) a mechanical translational system, (3) an incubation unit, (4) a control circuit, and (5) a controlling program. Each module is explained in detail below.

i. We used fibre-coupled partially coherent laser illumination (SC400-4, Fianium Ltd., Southampton, UK), with the wavelength and intensity controlled through an acousto-optic tunable filter (AOTF) device (Fianium Ltd., Southampton, UK). The device was remotely controlled with a customized program written in the C++ programming language and ran on a controlling laptop computer (product no. EON17-SLX, Origin PC). The laser light was transmitted through the sample, i.e., the agar plate that contains the bacterial colonies, and forms an inline hologram on a CMOS image sensor (product no. acA3800-14 μm, Basler AG, Ahrensburg, Germany) with a pixel size of 1.67 μm and an active area of 6.4 mm × 4.6 mm. The CMOS image sensor was connected to the same controlling laptop

原文设备模块明确给出1.67 μm像素和6.4×4.6 mm^2有效面积；单图原生覆盖约29.4 mm^2。

光经过样品后在传感器上形成同轴全息图。照明与扫描由程序同步，影像先缓冲再写盘。

模块实现提供的是测量结构，不是以设备清单替代分辨率、信噪比或回收率的验证。

原文截图 主文 PDF p.13, Methods：光学与扫描模块

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检测网络输入张量

DNN-enabled detection of growing bacterial colonies

Following the colony candidate selection process outlined earlier, we cropped out candidate regions of 160 × 160 pixels (~267 μm × 267 μm) across the four back-propagated consecutive frames and separated the complex field into amplitude and phase channels. Therefore, each candidate region is represented by a $2 \times 4 \times 160 \times 160$ array. This four-dimensional (phase/amplitude–time–x–y) data format differs from the traditional three-dimensional data used in image classification tasks and requires a custom-designed DNN architecture that accounts for the additional dimension of time. We designed our DNN by following the block diagram of DenseNet[28] and replaced the 2D convolutional layers with P3D convolutional layers[47], as shown in Supplementary Fig. S9. Our network was implemented in Python (v3.7.2) with the PyTorch Library (v1.0.1). The network was randomly initialized and optimized using an adaptive moment estimation (Adam) optimizer[48] with a starting learning rate of 1×10^{-4} and a

每个候选从四帧截取160×160像素，约267×267 μm²；分成幅度与相位两通道。

输入为2×4×160×160：通道×时间×x×y，共204,800个标量。

空间旋转/翻转用于增强，不改变时间顺序。该张量描述菌落邻域，并非单个细菌的直接图像。

原文截图 主文 PDF p.15, 检测网络输入、架构与训练

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分类网络输入与额外等待

4 h, and then are sent to the second DNN as a $2 \times 8 \times 288 \times 288$ array for the classification of colony species. To perform the classification task, this time, the training data only contain the true colonies and their corresponding species (ground truth). The network follows a similar structure and training process as the detection model, as illustrated in Supplementary Fig. S9. The network was randomly initialized and optimized using the Adam optimizer[48], with a starting learning rate of 1×10^{-4} and a batch size of 64. The learning rate decayed by 0.9 times every 10 epochs. To avoid overfitting to a specific plate, we discarded colony images extracted from extremely dense samples (>1000 CFU per plate). As a result, approximately 9400 growing colonies were used in the training and validation of the classification model. The whole training process took ~15 h using a desktop computer with dual GPUs (GTX1080Ti, Nvidia).

对已检出菌落再等2 h，积累8帧；输入为2×8×288×288。

较大空间邻域和较长时间窗可包含增长形态差异。作者排除了>1000 CFU/plate极密集板的分类训练图。

这会限制对拥挤、重叠菌落的外推；模型未因输入更大就自动具备未知菌种泛化。

原文截图 主文 PDF p.15, 分类输入、训练与排除高密度板

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Fig.S9｜两套P3D DenseNet

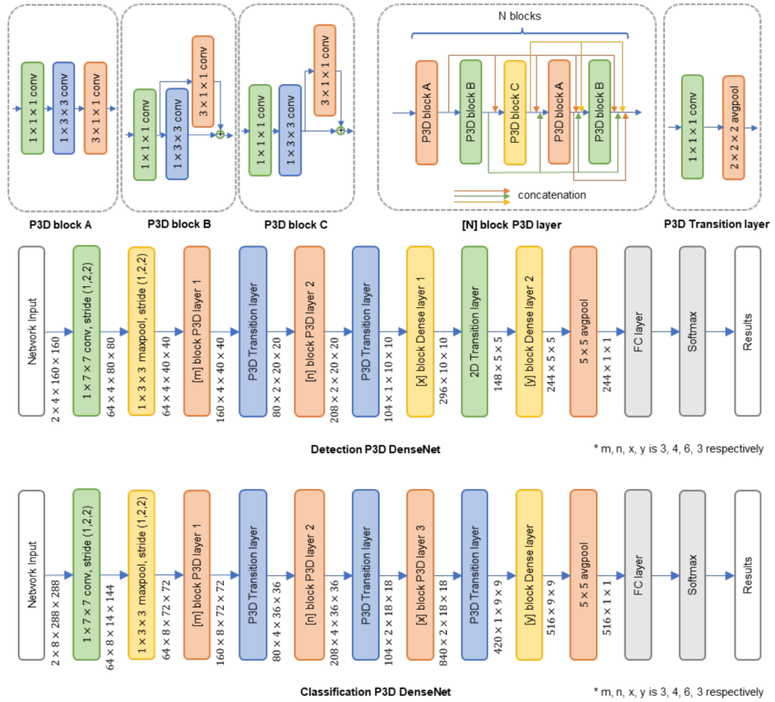


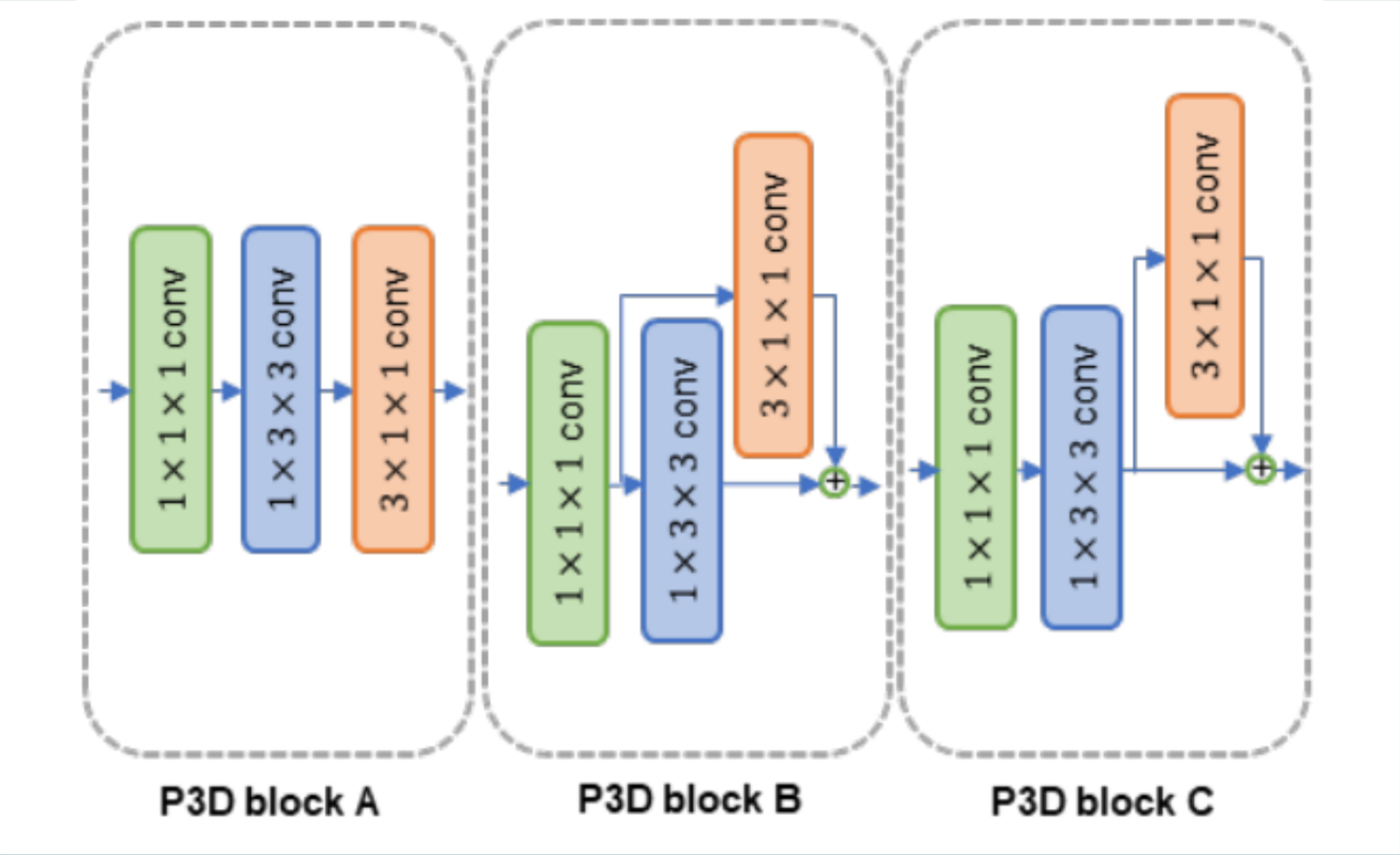
Figure S9. Schematic of pseudo-3D (P3D) DenseNet models for the detection and classification of growing colonies using the lens-free imaging system. The detection neural network model has 1.21×10^6 trainable parameters. The classification neural network model has 1.36×10^6 trainable parameters.

原文截图 补充 PDF p.10, Fig.S9 全图/表

上部是复用模块；中部检测网络；下部分类网络。参数量分别1.21×10^6与1.36×10^6。

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 补充 PDF p.10, Fig.S9 全图/表
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P3D模块如何利用时间



P3D将空间与时间卷积分开：1×3×3处理空间，3×1×1处理时间，1×1×1用于通道变换。

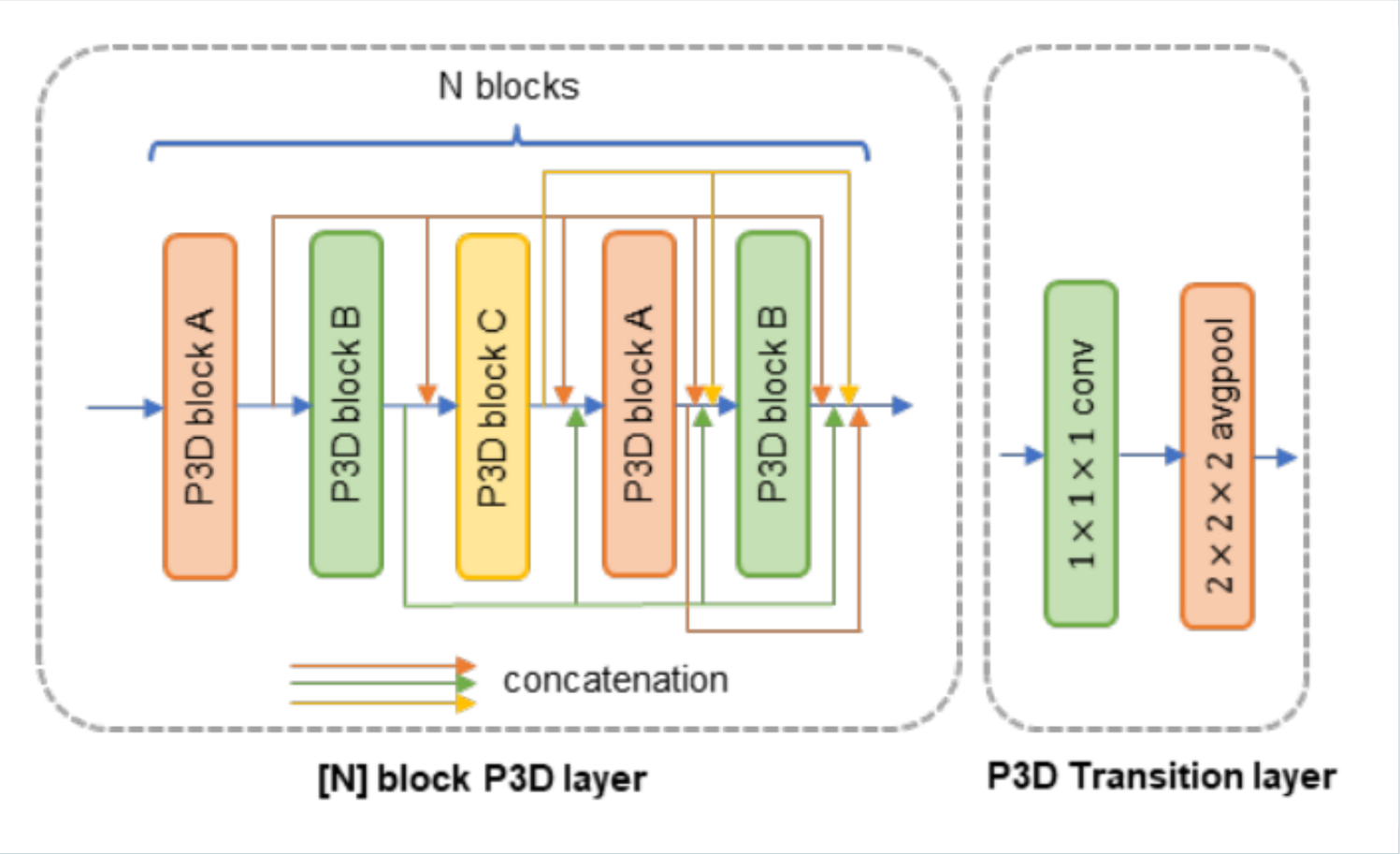
A空间后时间串联；B空间/时间并行相加；C将空间输出与其后时间卷积输出相加。加号为逐元素融合。

这是时空特征学习，不等同于手工计算“生长速率”。图仅给结构，不能据此声称获得可解释生物机制。

原文截图 补充 PDF p.10, Fig.S9 P3D blocks A/B/C

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 补充 PDF p.10, Fig.S9 P3D blocks A/B/C
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Dense连接与过渡层



Dense层把多个先前模块特征拼接到后续输入，图中彩色连接表示concatenation。

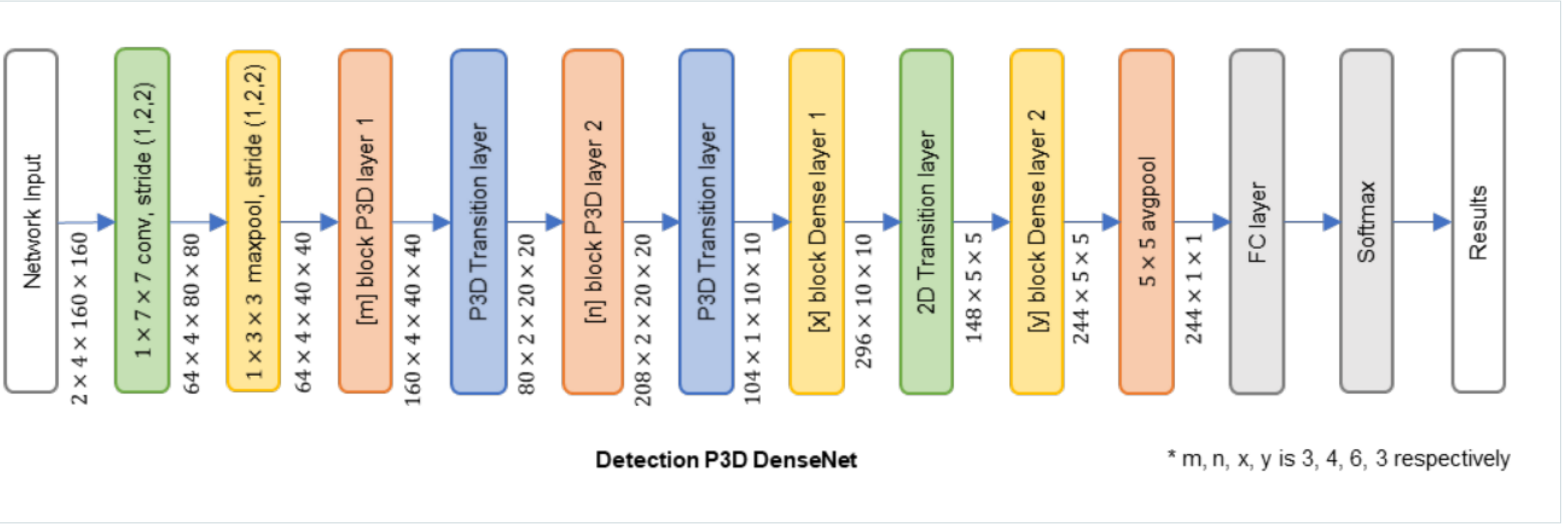
过渡层含1×1×1卷积和平均池化，改变通道量并压缩时空尺寸。

拼接与相加不是同一种操作：拼接增加通道维，相加要求对应张量形状一致。

原文截图 补充 PDF p.10, Fig.S9 dense连接与transition

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 补充 PDF p.10, Fig.S9 dense连接与transition
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检测网络｜完整张量流

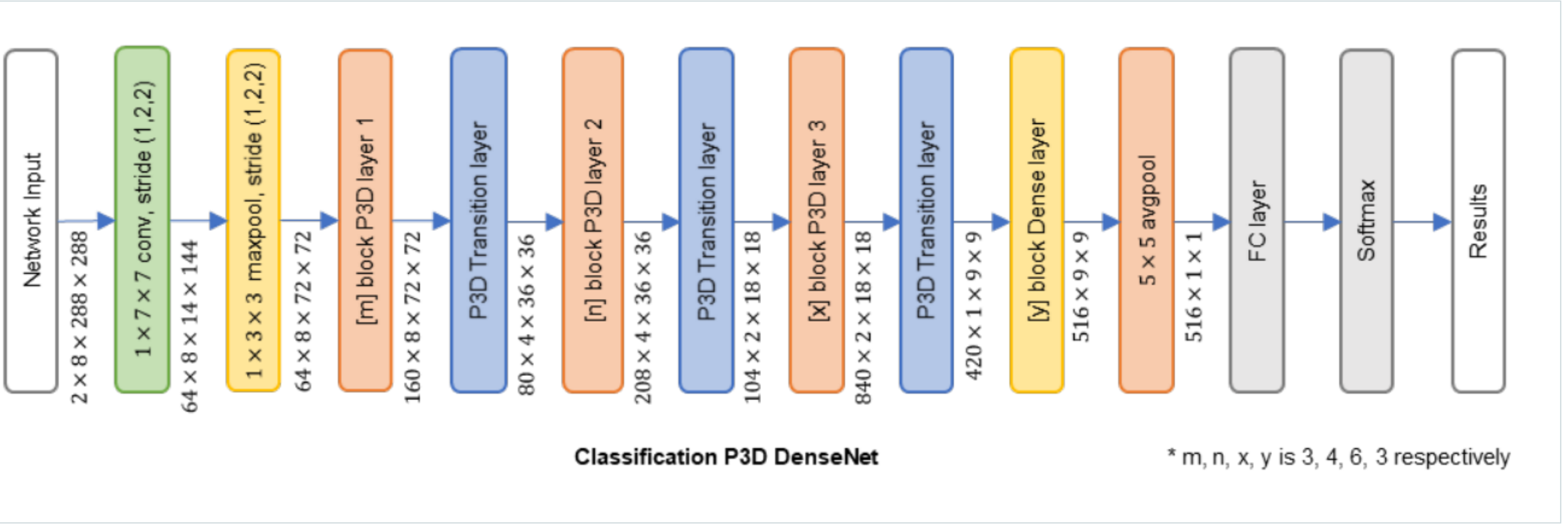


原文截图 补充 PDF p.10, Fig.S9检测网络完整链

由4帧输入，经P3D层/过渡层到全局平均池化、全连接及softmax。按原图保留每层尺寸。

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 补充 PDF p.10, Fig.S9检测网络完整链
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分类网络｜完整张量流

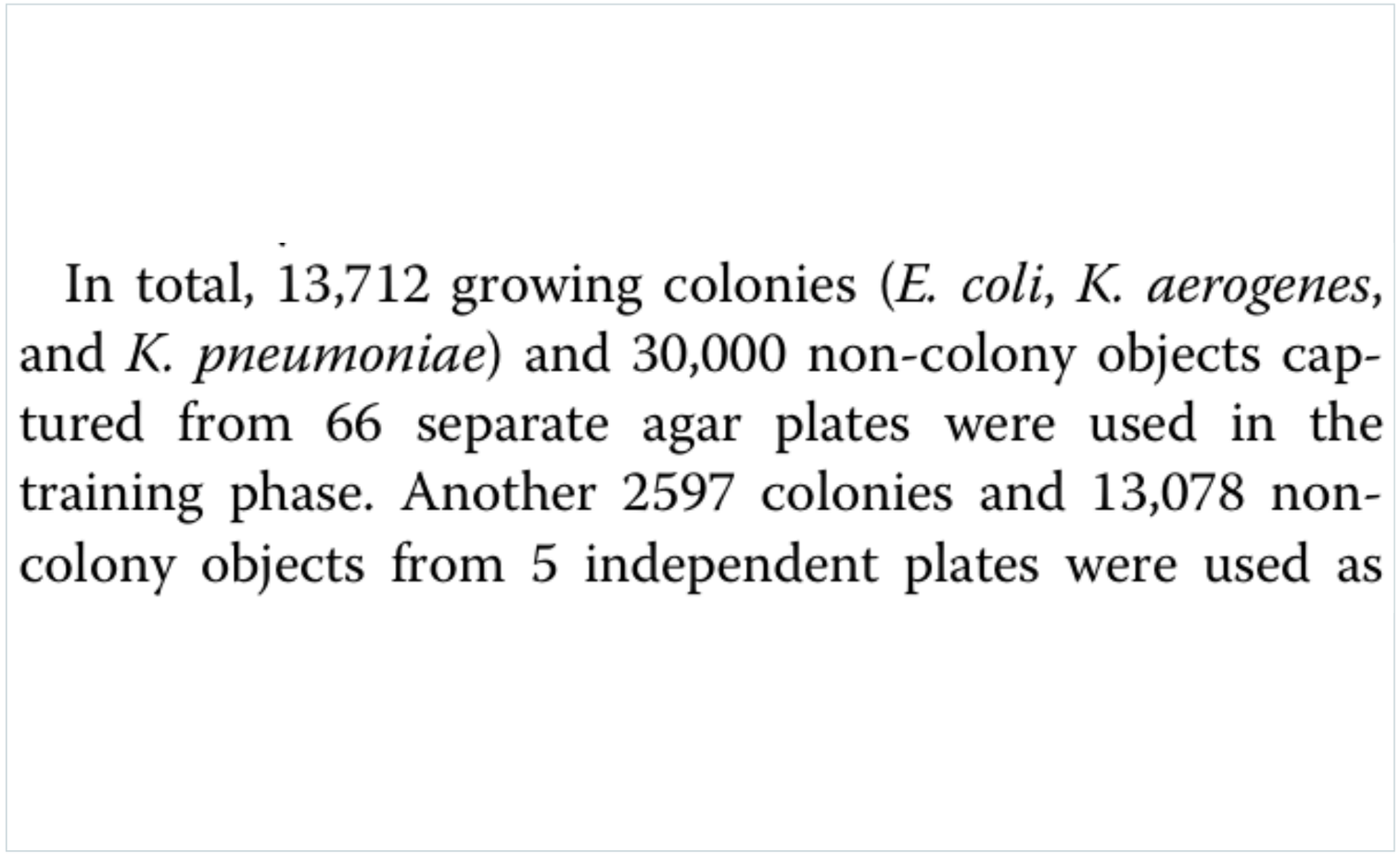


原文截图 补充 PDF p.10, Fig.S9分类网络完整链

8帧输入具有更多空间/时间样本；参数与层链独立于检测网络，不共享一个性能指标。

Wang et al. Light: Science & Applications 9, 118 (2020). DOI:10.1038/s41377-020-00358-9 | 补充 PDF p.10, Fig.S9分类网络完整链
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训练、验证与独立盲测



In total, 13,712 growing colonies (*E. coli*, *K. aerogenes*, and *K. pneumoniae*) and 30,000 non-colony objects captured from 66 separate agar plates were used in the training phase. Another 2597 colonies and 13,078 non-colony objects from 5 independent plates were used as

检测训练：66板、13,712菌落、30,000非菌落。验证：另5板、2,597菌落、13,078非菌落。

合计71板、16,309菌落与43,078非菌落，方法节以约16,000/43,000概述。

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原文截图 主文 PDF p.5, 检测训练/验证计数

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